Week 1 — Step-by-Step Solutions

Statistical Signal Processing · Tutorial 1

Course: Statistical Signal Processing (L.048.24014, Universität Paderborn, Dr.-Ing. Tanuj Hasija), WiSe 2026/27 **Covers:** Tutorial 1, Problems 1–5 **Checking:** no official solution was available when this was written. Every result is verified by `python/tutorial1_verify.py` (exact fractions, symbolic sums with SymPy, Monte Carlo simulation).

Every step is written out. References like "Concepts §6" point to `01_Concepts.md`.

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Problem 1 — Monty Hall

> Three doors: a car behind one, goats behind the other two. You pick door 3. The host, **who knows** where the car is, opens door 1, which has a goat, and offers you door 2. Should you switch?

Step 1 — State the rules of the game precisely

The answer depends on **how the host chooses**, so we make the standard assumptions explicit:

1. The car is placed uniformly at random: each door has probability $1/3$.
2. The host **always** opens a door that (a) you did not pick and (b) has a goat, and then **always** offers the switch.
3. If both unpicked doors have goats (i.e. your door has the car), the host chooses between them at random, each with probability $1/2$.

Step 2 — Causes, priors and the observation

Causes (a partition, Concepts §6.1): C_i = "car behind door i", $i = 1, 2, 3$, with priors

$$P(C_1) = P(C_2) = P(C_3) = \tfrac{1}{3}.$$

Observation: H_1 = "host opens door 1" (given that you hold door 3).

Step 3 — Likelihoods: how probable is "host opens door 1" under each cause?

- C_1 (car behind door 1): the host never reveals the car ⇒ $P(H_1 \mid C_1) = 0$.
- C_2 (car behind door 2): the host cannot open door 3 (yours) nor door 2 (car) ⇒ he **must** open door 1 ⇒ $P(H_1 \mid C_2) = 1$.
- C_3 (car behind your door 3): doors 1 and 2 both have goats, he picks one at random ⇒ $P(H_1 \mid C_3) = \frac{1}{2}$.

Step 4 — Total probability of the observation

$$P(H_1) = \sum_i P(H_1 \mid C_i)P(C_i) = 0 \cdot \tfrac{1}{3} + 1 \cdot \tfrac{1}{3} + \tfrac{1}{2} \cdot \tfrac{1}{3} = \tfrac{1}{2}.$$

Step 5 — Bayes' rule

$$P(C_2 \mid H_1) = \frac{P(H_1 \mid C_2)P(C_2)}{P(H_1)} = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}, \qquad P(C_3 \mid H_1) = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}, \qquad P(C_1 \mid H_1) = 0.$$

Check: $0 + \frac{2}{3} + \frac{1}{3} = 1$. ✓

Step 6 — Decision

$$\boxed{P(\text{win} \mid \text{switch}) = \tfrac{2}{3} \quad > \quad P(\text{win} \mid \text{stay}) = \tfrac{1}{3}}$$

Yes, you should switch. It doubles your chance of winning the car. A simulation of 200 000 games gives 0.668 for switching and 0.332 for staying.

Step 7 — Why the intuition "it's 50/50 now" is wrong

- **Counting argument:** your first pick is correct with probability $1/3$ and stays so: the host can **always** open a goat door, so his action tells you nothing about **your** door. Switching wins exactly when your first pick was wrong, which happens with probability $2/3$.
- **Information argument:** the host's choice is **not random**; it is constrained by where the car is. In case C_2 he is forced to open door 1, in case C_3 he only does so half the time. Door 1 being opened is therefore twice as likely under C_2 as under C_3, and the posterior shifts in favour of door 2 (likelihood ratio $1 : \frac{1}{2}$).
- **Exaggerated version:** with 100 doors, you pick one, the host opens 98 goat doors and leaves one closed. Almost surely the car is behind the door he carefully avoided (probability $99/100$).

Step 8 — The assumptions matter (connection to SSP)

If the host **did not know** where the car is and opened door 1 at random, which happened to show a goat, then $P(H_1 \cap \text{goat} \mid C_2) = P(H_1 \cap \text{goat} \mid C_3) = \frac{1}{2}$, and $P(\text{goat at door 1} \mid C_1) = 0$. Bayes then gives $P(C_2 \mid \cdot) = P(C_3 \mid \cdot) = \frac{1}{2}$: switching neither helps nor hurts. **The same observation carries different information depending on the mechanism that produced it.** In signal processing terms: the inference depends on the statistical model of the measurement, not just the measured value — the theme of the whole course.

Problem 2 — Poisson: hits on a website

Hits in a 5-minute interval ~ Poisson(4). Find P(at least one hit), P(at least 3 hits) and P(no more than 5 hits) between 3:00 and 3:05 pm.

Step 1 — Set up

X = number of hits between 3:00 and 3:05 pm. This is a 5-minute interval, so $X \sim$ Poisson($\lambda = 4$) (Concepts §10.4):

$$P_X(x) = \frac{4^x e^{-4}}{x!}, \qquad x = 0, 1, 2, ...$$

Step 2 — Tabulate the first few probabilities

Start with $P_X(0) = e^{-4}$ and use the ratio $P_X(x) = \frac{\lambda}{x} P_X(x-1)$:

x	$P_X(x)$ exact	multiple of e^{-4}	numerical
0	e^{-4}	1	0.018316
1	$4e^{-4}$	4	0.073263
2	$\frac{4^2}{2}e^{-4}$	8	0.146525
3	$\frac{4^3}{6}e^{-4}$	32/3	0.195367
4	$\frac{4^4}{24}e^{-4}$	32/3	0.195367
5	$\frac{4^5}{120}e^{-4}$	128/15	0.156293

Step 3 — At least one hit

"At least one" is the complement of "none" (Concepts §3.2):

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-4} \approx \boxed{0.9817}$$

Step 4 — At least three hits

The direct sum $\sum_{x=3}^{\infty}$ has infinitely many terms, so use the complement $\{X \leq 2\}$:

$$P(X \geq 3) = 1 - \left[P_X(0) + P_X(1) + P_X(2)\right] = 1 - e^{-4}(1 + 4 + 8) = 1 - 13e^{-4} \approx \boxed{0.7619}$$

Step 5 — No more than five hits

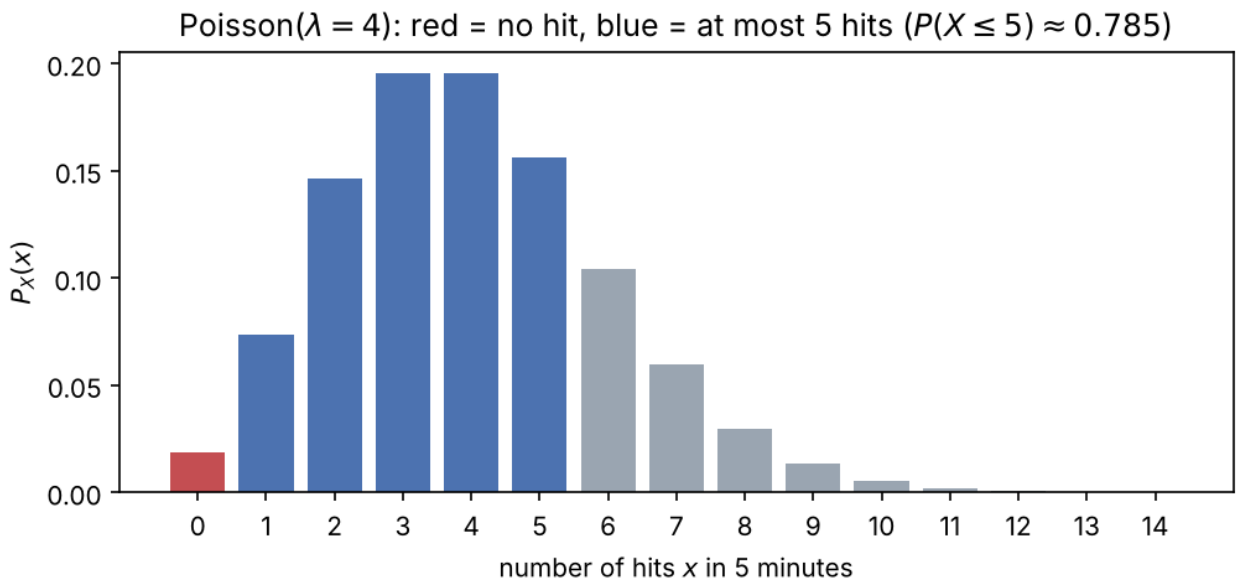
"No more than 5" means $X \leq 5$, a finite sum:

$$P(X \leq 5) = e^{-4}\left(1 + 4 + 8 + \tfrac{32}{3} + \tfrac{32}{3} + \tfrac{128}{15}\right) = \tfrac{643}{15}e^{-4} \approx \boxed{0.7851}$$

(Common denominator 15: $15 + 60 + 120 + 160 + 160 + 128 = 643$.)

Step 6 — Sanity checks

- The mean is 4, so "at least one hit" should be very likely (✓ 98%) and "at most 5" (a bit above the mean) should be well above 1/2 (✓ 79%).
- Simulation of 10^6 Poisson(4) samples: 0.9819, 0.7618, 0.7852. ✓



Problem 3 — Mean and variance of a Poisson RV

Obtain the mean and variance of $X \sim$ Poisson(λ).

Step 1 — The pmf sums to one (needed below)

$$\sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1.$$

Step 2 — Mean

By definition (Concepts §8.1):

$$E\{X\} = \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!}.$$

- The $x = 0$ term is zero, so start at $x = 1$. - For $x \geq 1$: $\frac{x}{x!} = \frac{1}{(x-1)!}$. - Pull out one factor λ: $\lambda^x = \lambda \cdot \lambda^{x-1}$.

$$E\{X\} = \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \overset{m=x-1}{=} \lambda e^{-\lambda} \sum_{m=0}^{\infty} \frac{\lambda^m}{m!} = \lambda e^{-\lambda} e^{\lambda}.$$

$$\boxed{E\{X\} = \mu_X = \lambda}$$

Step 3 — Second factorial moment $E\{X(X-1)\}$

Same trick, now with two factors (Concepts §8.4). The terms $x = 0$ and $x = 1$ vanish, and $\frac{x(x-1)}{x!} = \frac{1}{(x-2)!}$ for $x \geq 2$:

$$E\{X(X-1)\} = \lambda^2 e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} \overset{m=x-2}{=} \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2.$$

Step 4 — Variance

$E\{X^2\} = E\{X(X-1)\} + E\{X\} = \lambda^2 + \lambda$ (linearity), so

$$\sigma_X^2 = E\{X^2\} - \mu_X^2 = \lambda^2 + \lambda - \lambda^2 \;\Rightarrow\; \boxed{\text{var}(X) = \sigma_X^2 = \lambda}$$

Step 5 — Interpretation

- λ is the **average number of events** per interval (4 hits per 5 minutes in Problem 2).
- Mean = variance is a fingerprint of Poisson counts. The relative spread $\sigma_X/\mu_X = 1/\sqrt{\lambda}$ shrinks as λ grows: large counts are relatively more predictable. (This is the origin of "shot noise" in photodetectors.)
- Consistent with the Binomial limit (Concepts §10.4): Binomial$(n, \lambda/n)$ has mean λ and variance $\lambda(1-\lambda/n) \to \lambda$.

Problem 4 — Deriving the Binomial distribution

Derive $P_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$. Hint: $X = Y_1 + \cdots + Y_n$ with independent Bernoulli RVs Y_k.

Step 1 — Model

Each trial $k = 1, \ldots, n$ has an indicator

$$Y_k = \begin{cases} 1, & \text{success (probability } p), \\ 0, & \text{failure (probability } 1-p), \end{cases}$$

and the Y_k are independent. The number of successes is $X = Y_1 + \cdots + Y_n$, which takes values $x = 0, 1, \ldots, n$.

Step 2 — Write the event $\{X = x\}$ as a union of disjoint elementary events

An elementary outcome of the experiment is a 0/1 sequence $(y_1, \ldots, y_n)$. We have $X = x$ exactly when the sequence contains x ones and $n - x$ zeros:

$$\{X = x\} = \bigcup_{\substack{(y_1,\ldots,y_n)\in\{0,1\}^n \\ y_1+\cdots+y_n=x}} \{Y_1 = y_1, \ldots, Y_n = y_n\}.$$

Different sequences are different outcomes, so these events are **pairwise disjoint**, and by axiom 3 their probabilities add.

Step 3 — Probability of one particular sequence

By independence the joint probability factorises:

$$P(Y_1 = y_1, \dots, Y_n = y_n) = \prod_{k=1}^{n} P(Y_k = y_k) = \prod_{k=1}^{n} p^{y_k}(1-p)^{1-y_k} = p^{\sum_k y_k}(1-p)^{n-\sum_k y_k}.$$

For a sequence with x ones this is $p^x(1-p)^{n-x}$. **It does not depend on where the ones are**, only on how many there are. Example ($n = 3$, $x = 2$): $P(1,1,0) = P(1,0,1) = P(0,1,1) = p^2(1-p)$.

Step 4 — Count the sequences with exactly x ones

A sequence with x ones is fixed by choosing **which** x of the n positions hold the ones. The number of x-element subsets of n positions is $\binom{n}{x}$ (Concepts §4.2).

Step 5 — Add up

$$\boxed{P_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \dots, n}$$

(number of sequences) × (probability of each).

Step 6 — Checks

- **Normalisation:** binomial theorem, $\sum_{x=0}^{n} \binom{n}{x} p^x (1-p)^{n-x} = (p + (1-p))^n = 1$. ✓
- **Small case:** $n = 1$ gives Bernoulli; $n = 3$, $p = 1/2$ gives $1/8, 3/8, 3/8, 1/8$, the coin table from the lecture. ✓
- **Moments for free:** $E\{X\} = \sum_k E\{Y_k\} = np$ and, by independence, $\text{var}(X) = \sum_k \text{var}(Y_k) = np(1-p)$.
- The script enumerates all 2^5 sequences for $n = 5$, $p = 3/10$ and reproduces the formula exactly.

Alternative derivation — induction with the convolution formula

Let $X_n = Y_1 + \cdots + Y_n$. Then $X_n = X_{n-1} + Y_n$ with Y_n independent of X_{n-1}. By total probability over Y_n:

$$P(X_n = x) = P(X_{n-1} = x)(1-p) + P(X_{n-1} = x-1)\, p.$$

Assume $P(X_{n-1} = k) = \binom{n-1}{k} p^k (1-p)^{n-1-k}$. Then

$$P(X_n = x) = p^x (1-p)^{n-x} \left[\binom{n-1}{x} + \binom{n-1}{x-1} \right] = \binom{n}{x} p^x (1-p)^{n-x}$$

by Pascal's rule. The base case $n = 1$ is the Bernoulli pmf. ✓

Problem 5 — Channel decoder and retransmissions

8-bit codewords; bit errors independent with probability p. Fewer than 3 errors: corrected. Exactly 3: detected, retransmission requested (retransmissions independent). More than 3: neither corrected nor detected, decoded wrongly. Find (a) P(retransmission), (b) P(exactly n_R retransmissions), (c) the mean number of retransmissions.

Step 1 — Model one transmission

Let K = number of erroneous bits in one transmitted codeword. Each of the 8 bits is in error independently with probability p, so by Problem 4

$$K \sim \text{Binomial}(8, p), \qquad P(K = k) = \binom{8}{k} p^k (1-p)^{8-k}, \quad k = 0, \dots, 8.$$

The decoder's action depends only on K:

K	decoder action	probability
$0, 1, 2$	correct the errors → **correct** codeword	$P_c = \sum_{k=0}^{2} \binom{8}{k} p^k (1-p)^{8-k}$
3	detect, cannot correct → **retransmission**	$q = \binom{8}{3} p^3 (1-p)^5$
$4, \dots, 8$	neither → **wrong** codeword accepted	$P_w = 1 - P_c - q$

Step 2 — (a) Probability of retransmission

A retransmission is requested exactly when $K = 3$. With $\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$:

$$\boxed{q = P(\text{retransmission}) = P(K = 3) = 56\, p^3 (1-p)^5}$$

Step 3 — (b) Probability of exactly n_R retransmissions

Let N_R be the number of retransmissions for one codeword. The procedure is:

- Transmission 1: with probability q a retransmission is requested; with probability $1 - q$ the procedure **ends** (codeword corrected, or wrongly accepted — either way no retransmission).
- Each retransmission is again a fresh, **independent** transmission with the same probabilities.

So $N_R = n_R$ means: the first n_R transmissions all ended in "retransmit" and transmission number $n_R + 1$ did not. By independence (multiply along the chain, Concepts §5.1):

$$\boxed{P(N_R = n_R) = q^{n_R}(1-q), \qquad n_R = 0, 1, 2, \dots, \qquad q = 56 p^3 (1-p)^5}$$

This is the **Geometric** distribution (Concepts §10.3). Check: $\sum_{n_R \geq 0} q^{n_R}(1-q) = (1-q) \cdot \frac{1}{1-q} = 1$ ✓ (geometric series, valid since $q < 1$).

Step 4 — (c) Mean number of retransmissions

$$E\{N_R\} = \sum_{n_R=0}^{\infty} n_R\, q^{n_R}(1-q) = (1-q) \sum_{n_R=0}^{\infty} n_R\, q^{n_R} = (1-q) \frac{q}{(1-q)^2}$$

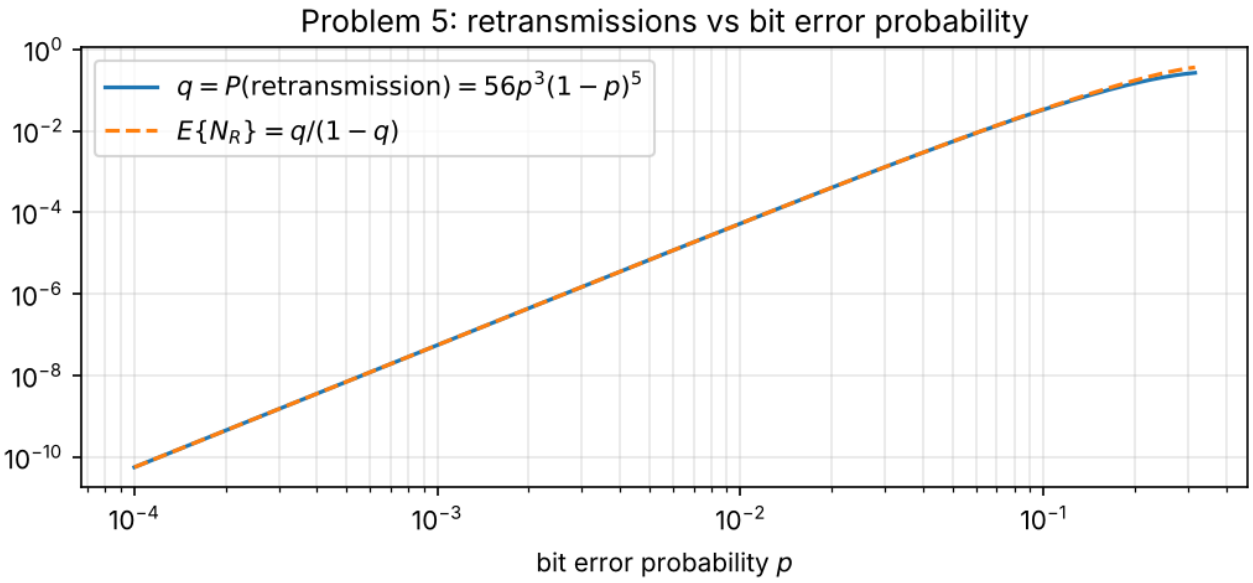
using $\sum_{n\geq 0} nq^n = q/(1-q)^2$ (derivative of the geometric series, Concepts §11.2):

$$\boxed{E\{N_R\} = \frac{q}{1-q} = \frac{56p^3(1-p)^5}{1-56p^3(1-p)^5}}$$

Step 5 — Numbers and interpretation

p	q	$E\{N_R\}$	P_c (per transmission)	P_w (per transmission)
10^{-3}	5.57×10^{-8}	5.57×10^{-8}	≈ 1	7.0×10^{-11}
10^{-2}	5.33×10^{-5}	5.33×10^{-5}	0.999946	6.8×10^{-7}
10^{-1}	3.31×10^{-2}	3.42×10^{-2}	0.9619	5.0×10^{-3}

- For small p, $(1-p)^5 \approx 1$ and $q \approx 56p^3$, which is tiny: retransmissions are rare, and $E\{N_R\} \approx q$.
- $E\{N_R\}$ grows quickly with p (as p^3 for small p), see the figure.
- The simulation of 200 000 codewords matches the formula (e.g. 0.0345 vs 0.0342 at $p = 0.1$).
- **Bonus:** the probability that the codeword is **eventually** decoded correctly (after any number of retransmissions) is, by the same geometric argument, $\sum_{n\geq 0} q^n P_c = \frac{P_c}{1-q}$, and the probability of eventually accepting a wrong codeword is $\frac{P_w}{1-q}$. Retransmission turns "detected" errors into a fresh chance, but cannot help with the undetected ones.



Summary of results

Problem	Result
1 Monty Hall	switch: $P(\text{win}) = 2/3$; stay: $1/3$ ⇒ **switch**

Problem	Result
2 Poisson(4)	$P(X \geq 1) = 1 - e^{-4} \approx 0.9817$; $P(X \geq 3) = 1 - 13e^{-4} \approx 0.7619$; $P(X \leq 5) = \frac{643}{15}e^{-4} \approx 0.7851$
3 Poisson moments	$E\{X\} = \lambda$, var$(X) = \lambda$
4 Binomial	$P_X(x) = \binom{n}{x}p^x(1-p)^{n-x}$ = (#sequences) × (probability of one)
5a	$q = 56p^3(1-p)^5$
5b	$P(N_R = n_R) = q^{n_R}(1-q)$, $n_R = 0, 1, ...$ (geometric)
5c	$E\{N_R\} = q/(1-q)$